AYISHIK DAS

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Education

- **Academy of Technology (Under MAKAUT/WBUT)** Kolkata, India
  Bachelor of Technology - Computer Science Engineering and Business Systems ; CGPA: 8.28 Oct 2021 - June 2025
  Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Databases, Artificial Intelligence, Machine Learning

Skills Summary

- **Languages** C++, C, Python, Java, JavaScript, Shell
- **Systems/Performance** Data Structures, Algorithms, SIMD, HNSW
- **Backend/Databases** Node.js, Express.js, FastAPI, PostgreSQL, MySQL, SQLite, Redis
- **Frontend/AI** React.js, Next.js, Generative AI, RAG, Vector Search
- **Tools** Git, GitHub, Docker, Linux, WSL, Make, Postman

Experience

- **System Engineer** Kolkata, India
  Tata Consultancy Services Jan 2026 – Present
  - Engineered **3 Generative AI PoCs** for enterprise clients, including **2 solutions** involving conversational AI, workflow automation, and enterprise data integration.
  - Developed AI-driven solutions using **Generative AI, Claude, and Python libraries**, integrating with **PeopleSoft** to automate enterprise workflows and data processing.
  - Automated targeted enterprise workflows, reducing manual processing time by approximately **70%** and improving operational efficiency.

Projects

- **FluxTrade (C++20, CMake, Lock-Free Queues, Order Book, Matching Engine)** Built a deterministic electronic exchange simulator in **Modern C++20** with preallocated object pools, intrusive data structures, cache-line optimization, and lock-free queues. Implemented a **price-time priority limit order book and matching engine** with zero heap allocation on the critical execution path. Benchmarked **68.9 ns single-trade matching** and **161.5 ns multi-level sweeps** in an LTO Release build.
  GitHub: FluxTrade *July 2026 – Present*
- **Recallix (C++, Python, SQLite, HNSW, Generative AI, RAG)** Built a high-performance persistent memory engine for AI agents with **SIMD-accelerated HNSW indexing**, supporting **1M memory nodes** with **0.106ms average C++ search latency**. Implemented asynchronous index construction, SQLite WAL-based persistence, hybrid neural-heuristic reranking, and cross-model memory transfer achieving **100% retrieval fidelity**.
  GitHub: Recallix **Live:** recallix.vercel.app *Jan 2026 – Apr 2026*
- **Vault (C++, TCP, RESP, CMake, Redis Protocol)** Built a lightweight **Redis-compatible versioned key-value store** in C++, supporting RESP protocol communication through `redis-cli`. Implemented **hash-map based storage, snapshot versioning, custom RESP parsing, and TCP client-server communication**, with support for `COMMIT`, `CHECKOUT`, `DIFF`, and `RESET` operations.
  GitHub: Vault *May 2025 – June 2025*
- **PayP (Node.js, PostgreSQL, Sequelize, Express.js, JWT, bcrypt)** Designed and built a **UPI-inspired payment API** exposing reusable endpoints for user authentication, balance management, payment transfers, transaction history, and project/product payments. Implemented **JWT authentication, bcrypt credential hashing, payment PIN protection, and 30-minute account lockouts** against repeated authentication failures.
  GitHub: PayP **Live:** payp.onrender.com *Jan 2025 – May 2025*

Awards and Certifications

- Secured **2nd Place** in **TCS AI Friday**, Kolkata
- Winner of **TechQuisitive 5.0 (Quiz Contest)** and **CxthSense 2.0 (Coding Contest)** at College
- Achieved **5 star ratings in C++ and Java** on **HackerRank**
- Open-source Mentor at **GirlScript Summer of Code** and **Winter of Blockchain**
- **Postman API Fundamentals Student Expert Certificate**
- **Linux Command Line: From Zero to Hero Certificate**

Volunteer Experience

- **Member of IEI Student's Chapter CSBS** Aug 2023-May 2025
  Organizing and hosting different events in college
- **Event Coordinator - TechFiesta[Tech Fest of AOT]** Apr 2024
  Hosted **TechFest of AOT** and **event coordinator of Maths** event.